

TRANSPLANTATION

Extended HLA haplotypes and transplant survival

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KEY POINTS

- Extended HLA haplotypes are proxies for SNPs that affect relapse and mortality after haploidentical transplantation.
- Prospective selection of haploidentical donors can be optimized with knowledge of their extended HLA haplotypes.

The benefit of extended phased (~) HLA class I~class III~class II haplotypes in reducing the mortality after hematopoietic cell transplantation is unknown and requires information on functional class III variation. We identified a robust class III single-nucleotide polymorphism (SNP), rs915654, informative for mortality and relapse in 1436 patients and their haploidentical related donors through multivariable regression analysis of 26 candidate class III SNPs. Three-marker haplotypes, as defined by 1 class I locus, 1 class II locus, and rs915654 were determined in patients separately from donors. Inclusion of rs915654 into relapse and mortality models already containing patient *HLA-E~DRB1* and donor *HLA-B~DRB1* improved each model (likelihood ratio test $P = .06$ and $P = .004$, respectively, for relapse; $P = .10$ and $P = .01$, respectively, for mortality). The risks of mortality and relapse increased with decreasing numbers of favorable patient and donor markers. Retesting in an independent cohort of 1141 haploidentical transplants yielded similar results. The number of unfavorable markers additionally increased nonrelapse mortality. *HLA-A~C~B~DRB1~DQB1* haplotypes were defined according to their

expected numbers of favorable markers, and the theoretical utility for selecting donors was explored. In summary, extended HLA class I~class III~class II haplotypes influence the success of transplantation and inform the biology of the major histocompatibility complex in health and disease. The selection of haploidentical donors for future patients may be optimized with knowledge of donor HLA haplotypes.

Introduction

The human major histocompatibility complex (MHC) is home to >300 genes that participate in the immune response. Class I (HLA-A, -C, and -B) and class II (HLA-DR, -DQ, and -DP) molecules interface with T and natural killer cells and are well-defined determinants of hematopoietic cell transplant (HCT) outcome. Class I and II are bridged by class III, the largest, most gene-dense region spanning 900 kilobases with 75 loci (Figure 1).¹ Class III gene products participate in stress and inflammatory responses and the complement cascade and have well-established roles in infection, autoimmunity, cancer, and pharmacogenomics.¹⁻¹³ Their significance in HCT remains undefined.

HLA-associated susceptibility to diseases may be conferred by individual genes or by haplotypes of ≥ 2 physically linked markers that are coinherit through classical Mendelian segregation.^{14,15} *HLA-A, -C, -B, -DRB1, -DQB1, and -DPB1* define HLA haplotypes that are proxies for untyped variation and are powerful tools for mapping genes that cause disease.¹⁶ A hallmark of the MHC is the long-range and strong positive linkage disequilibrium (LD) between loci that can

extend over vast (kilobases) genetic distances. Phased (~) *HLA-A1~C7~B8~DR3~DQ2* and *-A3~C7~B7~DR15~DQ6* haplotypes are 2 well-known extended haplotypes that are strongly associated with autoimmunity and have earned them the name “inflammatory” haplotypes.^{2,6,7}

Patients and haploidentical family members share 1 identical haplotype and differ in the second. In sharp contrast to unrelated donor transplantation, wherein more complete and precise donor HLA matching is associated with improved outcomes with calcineurin inhibitor–based graft-versus-host disease prophylaxis,¹⁷ the risk of relapse after haploidentical transplantation is influenced by (mis)matching at individual HLA loci, by the presence or absence of specific sequence polymorphisms, and by additive effects of loci encoded as class II haplotypes.¹⁷⁻²¹ Hence, an understanding of HLA alloreactivity in relapse after transplantation must include consideration of germ line effects as well as patient-donor matching. A paradigm of haplotype-linked germ line variation does not require that the same loci are functional in both the patient and the donor because the model is not based on patient-donor mismatching.¹⁹ The significance of HLA haplotypes is brought into perspective when confronting the chief

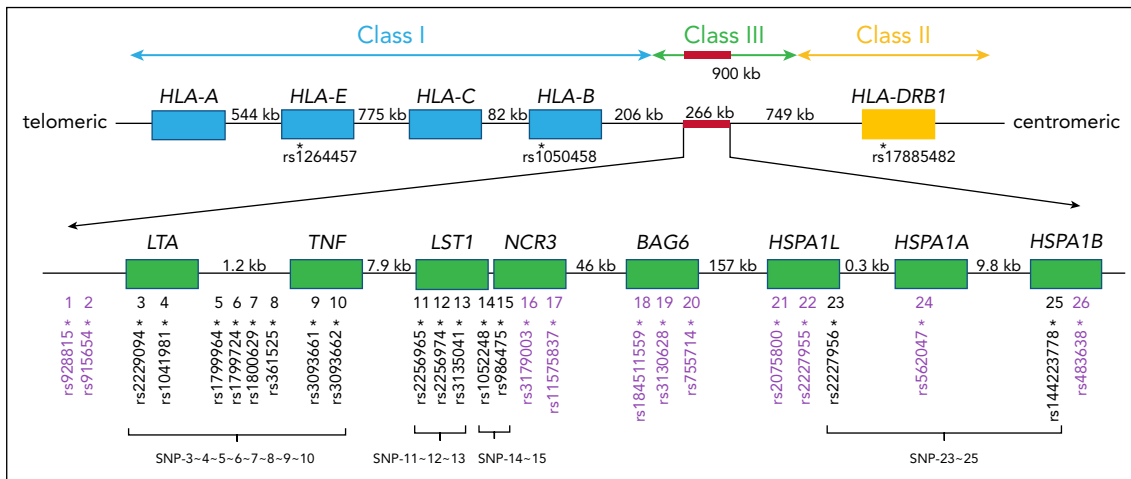


Figure 1. Map of the MHC. The MHC is a multigene complex, residence of classical *HLA-A*, *-C*, and *-B* (blue), and *-DRB1* (yellow) genes that each have important immunological function in HCT. This study sought to determine the clinical significance of variation within the class III region (green genes), focusing on 26 SNPs within a 266-kilobase region (red). The identification of a functional class III SNP permitted extended haplotypes composed of combinations of *HLA-E-Gly107Arg* (rs1264457), *HLA-B-Met-21Thr* (rs1050458), and *HLA-DRB1-Val86Gly* (rs17885482) to be evaluated for associations with clinical outcomes. Each of the 26 class III SNPs is identified by its rs name and by a surrogate number from 1 to 26 (telomere to centromere). The mapping of class III SNPs was facilitated by the analysis of SNP-haplotype “blocks” representing discrete regions of high LD. A total of 4 SNP-haplotype blocks (indicated by brackets and bold font) were identified in the global 1000 Genomes populations (details for LD blocks can be found in the supplemental Methods). Each block was tested for association with the clinical outcome. Of the 26 SNPs, 11 reside outside of blocks (purple font) and were individually assessed for association with clinical outcome. LTA, lymphotoxin alpha; TNF, tumor necrosis factor; LST1, leukocyte specific transcript 1; NCR3, natural cytotoxicity triggering receptor 3; BAG6, BAG chaperone 6; HSPA1L, heat shock protein family A (Hsp70) member 1 like; HSPA1A, heat shock protein family A (Hsp70) member 1A; HSPA1B, heat shock protein family A (Hsp70) member 1B.

unmet need in HCT today: to understand the underlying biology of disease recurrence so that new approaches can be developed to lower risks. We tested the hypothesis that extended patient and donor HLA haplotypes spanning class I, II, and III inform survival and relapse after haploidentical HCT. These novel data permitted the development of a new approach for optimizing the selection of haploidentical donors for future patients.

Methods

Patients and donors

A total of 2577 patients underwent haploidentical HCT with posttransplant cyclophosphamide (PTCy) in the United States for whom clinical data were reported to the Center for International Blood and Marrow Transplant Research (CIBMTR), and research biospecimens were available from the CIBMTR Biorepository. The clinical significance of single-nucleotide polymorphisms (SNPs) and SNP haplotypes was studied in 1436 transplants (cohort I; discovery cohort) and the consistency of findings assessed in 1141 independent transplants (cohort II; validation cohort; Table 1). The class III variants under study were not considered at the time of transplantation. The hypotheses were formulated independent of data collection and before genotyping. There were no exclusion criteria.

To explore a haplotype-based approach for donor selection, the 2577 transplants were supplemented with 193 patients who underwent haploidentical HCT without PTCy during the same time period for whom biospecimens were available, providing >8100 haplotypes for frequency calculations. Protocols were approved by the institutional review boards of the National Institutes of Health Office for Human Research Protections, Fred

Hutchinson Cancer Center, and the National Marrow Donor Program. Written informed consent was obtained for participation in the CIBMTR Research Database and Biorepository.

HLA and SNPs

HLA-A, *-C*, *-B*, *-DRB1*, and *-DQB1* were typed before transplant at 2-field resolution.²¹ *HLA-DPB1* typing was unavailable for a significant number of transplants. Patient and donor DNAs were characterized for 26 class III region SNPs (SNP1 through SNP26; Figure 1; supplemental Methods; supplemental Table 1, available on the Blood website). To define risks associated with *HLA-A~B~DRB1* haplotypes, analyses leveraged SNP-haplotype block structure assessed using standard measures of LD (supplemental Methods; supplemental Figure 1) to first identify class III SNPs associated with outcome by testing strings of genotypes and phased haplotypes within blocks. If blocks were associated with outcome, each intrablock SNP was tested (described in “Biostatistical methods”). SNPs residing outside of blocks were individually assessed for associations with the outcome. Mismatching was defined in either graft-versus-host or host-versus-graft vectors and tested in blocks and single SNPs. This approach yielded candidate SNPs in patients and donors that merited testing in class I~class II foundational haplotypes.

Class I~class II foundational haplotypes were identified by testing combinations of 1 class I with 1 class II marker each previously shown to influence mortality and relapse in haploidentical HCT: *HLA-E-Gly107Arg* (rs1264457; residue 107 Gly→Arg dimorphism), *HLA-B*-leader position 2 Met/Thr (rs1050458; henceforth “*HLA-B*-Met-21Thr”), and *HLA-DRB1-Val86Gly* (rs17885482; residue 86 Val→Gly dimorphism).^{17-19,21}

Table 1. Demographics of the study population

Characteristic	Cohort I patients and donors n = 1436 pairs	Cohort II patients and donors n = 1141 pairs
Recipient age (y), n (%)		
Mean (SD)	47.2 (19.1)	46.2 (21.4)
Median (minimum, maximum)	52.3 (0.72, 78.4)	52.7 (0.5, 82.0)
Donor age (y), n (%)		
Mean (SD)	37.3 (13.2)	34.8 (12.6)
Median (minimum, maximum)	36.2 (3.9, 76.9)	34.3 (2.4, 73.2)
Not specified	1 (<1)	0 (0)
Year of transplantation, n (%)		
Mean (SD)	2016 (1.90)	2021 (1.7)
Median (minimum, maximum)	2017 (2009, 2019)	2021 (2010, 2023)
Patient sex, n (%)		
Female	573 (40)	481 (42)
Male	863 (60)	660 (58)
Donor sex, n (%)		
Female	559 (39)	468 (41)
Male	877 (61)	673 (59)
Relationship of donor to patient, n (%)		
Child	685 (48)	567 (50)
Sibling	473 (33)	317 (28)
Parent	127 (9)	187 (16)
Other	151 (11)	70 (6)
Disease, n (%)*		
ALL	240 (17)	278 (24)
AML	573 (40)	511 (45)
CML	53 (4)	19 (2)
Lymphoma	151 (11)	38 (3)
MDS/MPN	263 (18)	249 (22)
Myeloma	29 (2)	5 (<1)
Other malignant blood disorders	37 (3)	12 (1)
Other malignant nonblood disorders	2 (<1)	0 (0)
Nonmalignant†	88 (6)	29 (3)
Disease risk index, n (%)‡		
Low	114 (8)	68 (6)
Intermediate	638 (44)	494 (43)
High	313 (22)	242 (21)
Very high	53 (4)	17 (1)
Not specified	318 (22)	320 (28)
Total body irradiation, n (%)		
Yes	995 (69)	811 (71)
No	441 (31)	330 (29)

ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CML, chronic myeloid leukemia; MAC, myeloablative conditioning; MDS, myelodysplastic syndrome; MPN, myeloproliferative disorder. NMA, nonmyeloablative; RIC, reduced-intensity conditioning; SCID, severe combined immunodeficiency; SD, standard deviation.

*Lymphoma, including Hodgkin and non-Hodgkin lymphoma, other malignant blood disorders (leukemia not specified), and other malignant nonblood disorders (solid malignancies).

†Nonmalignant (autoimmune diseases, histiocytic disorders, inherited abnormality of erythrocyte differentiation or function, inherited disorder of metabolism, severe aplastic anemia, SCID, and other immune system disorders).

‡Disease risk index.²²

Table 1 (continued)

Characteristic	Cohort I patients and donors n = 1436 pairs	Cohort II patients and donors n = 1141 pairs
Preparatory regimen intensity, n (%)		
MAC	596 (42)	557 (49)
RIC/NMA	733 (51)	583 (51)
Not specified	107 (7)	1 (<1)
Source of cells, n (%)		
Bone marrow	661 (46)	329 (29)
Peripheral blood stem cells	772 (54)	812 (71)
Bone marrow and peripheral blood stem cells	3 (<1)	0 (0)
Patient/donor cytomegalovirus serological status, n (%)		
Negative/negative	277 (19)	221 (19)
Negative/positive	390 (27)	114 (10)
Positive/negative	159 (11)	324 (28)
Positive/positive	599 (42)	476 (42)
Not specified	11 (1)	6 (1)
PTCy, n (%)		
Yes	1436 (100)	1141 (100)
Patient ancestry, n (%)		
Asian	73 (5)	64 (6)
Black	298 (21)	163 (14)
White	755 (53)	580 (51)
Hispanic	190 (13)	273 (24)
Other/not specified	120 (8)	61 (5)
HCT-specific comorbidity index, n (%)		
0	294 (20)	227 (20)
1	196 (14)	184 (16)
2	230 (16)	176 (15)
≥3	705 (49)	545 (48)
Not specified	11 (1)	9 (1)
HLA-A, -B, -C, -DRB1, and -DQB1 match grade, n (%)		
5/10	849 (59)	736 (65)
6/10	287 (20)	222 (19)
7/10	111 (8)	84 (7)
8/10	32 (2)	20 (2)
9/10	18 (1)	3 (<1)
10/10	6 (<1)	0 (0)
Not specified	133 (9)	76 (7)

ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; CML, chronic myeloid leukemia; MAC, myeloablative conditioning; MDS, myelodysplastic syndrome; MPN, myeloproliferative disorder. NMA, nonmyeloablative; RIC, reduced-intensity conditioning; SCID, severe combined immunodeficiency; SD, standard deviation.

*Lymphoma, including Hodgkin and non-Hodgkin lymphoma, other malignant blood disorders (leukemia not specified), and other malignant nonblood disorders (solid malignancies).

†Nonmalignant (autoimmune diseases, histiocytic disorders, inherited abnormality of erythrocyte differentiation or function, inherited disorder of metabolism, severe aplastic anemia, SCID, and other immune system disorders).

‡Disease risk index.²²

Each candidate class III SNP was tested to identify the SNP that most improved the fit for mortality and relapse models beyond the foundational haplotype for construction of class I~class III~class II haplotypes.

Phase (~) for *HLA-A~C~B~DRB1~DQB1* was established through familial segregation (supplemental Methods). Phase for the 26 class III region SNPs was determined by homozygosity and shared/nonshared polymorphisms between the

patient and haploidentical donor. *HLA-B*-Met-21Thr~*DRB1*-Val86Gly phase was established through knowledge of the (mis)matched *HLA-B* and *HLA-DRB1* alleles of the patient and the haploidentical donor's shared and nonshared HLA haplotypes. Phase for *HLA-E*-Gly107Arg~rs915654 (SNP2) was established through homozygosity and knowledge of shared/nonshared polymorphisms.

Biostatistical methods

This study evaluated the overarching hypothesis that extended HLA haplotypes are associated with overall mortality (studied in all patients) and relapse of underlying malignancy (studied only in patients with a malignant disorder) after haploidentical transplantation using PTCy and was established at the outset of the study. We used a classic discovery-validation study design to identify putative functional class III markers in cohort I (1436 transplants) and assess candidate SNPs for consistency of findings in cohort II (1141 transplants). Cohort I and cohort II were defined according to the chronological availability of research biospecimens. Haplotypes were constructed using markers known to influence mortality and relapse,^{18,19,21} but the effect of haplotypes on nonrelapse mortality was also investigated. The associations of genotypes, haplotypes, and patient/donor SNP mismatching with the cause-specific hazards of relapse and the hazards of mortality were assessed using Cox regression. Cox regression was chosen for relapse because we were interested in assessing the underlying biological effect of the studied genetic variants on the outcome.²³ Potential confounding variables for models included patient age; donor age; patient sex; donor sex; year of transplant; intensity of conditioning regimen (myeloablative and reduced-intensity/nonmyeloablative); disease risk index (low, intermediate, high, and very high)^{22,24}; patient/donor cytomegalovirus serostatus; source of stem cells (bone marrow vs peripheral blood); use of total body irradiation; comorbidity index (0, 1, 2, and ≥ 3); patient race; family relationship of donor; and *HLA-A*, *-C*, *-B*, *-DRB1*, and *-DQB1* match status as appropriate.²¹ If a categorical covariate was missing data, an additional category was created to reflect the missing value of the appropriate covariate. The Wald test was used to obtain 2-sided *P* values from Cox regression models. The analysis used a discovery-validation study design, and no adjustments for multiple comparisons were made,^{25,26} but a global test was conducted for comparisons involving >2 groups; if the global *P* value was $>.05$, pairwise *P* values were not presented. A global *P* value was used to determine whether a block had an association with the clinical end point, and values of $\leq .10$ were considered of interest for testing of single SNPs within the block.

For haplotype construction, single class III intrablock SNPs that did not have ≥ 50 patients/donors in all genotype groups were excluded from consideration of inclusion into class I~class II foundational haplotypes. To define the clinical relevance of the number of (un)favorable SNPs, an unfavorable SNP allele was defined if the hazard ratios of mortality and relapse relative to the alternate allele were >1.0 ; otherwise, the allele was defined as favorable. The association between the number of (un)favorable SNPs and outcome (mortality and relapse) was assessed using a trend test in which the number of (un)favorable SNPs was modeled as a continuous linear variable (with integer values of 0-6), and this value was regressed (with Cox

regression) on outcome. The total expected number of favorable SNPs for each *HLA-A*~*B*~*DRB1* haplotype was estimated from all 2770 transplants regardless of the use of PTCy. The expected number (also interpreted as an estimate of the mean number in the grand population) was calculated as the sum of the (frequency of 0 favorable SNPs $\times$ 0) + (frequency of 1 favorable SNP $\times$ 1) + (frequency of 2 favorable SNPs $\times$ 2) + (frequency of 3 favorable SNPs $\times$ 3), that is, a weighted average of the number of favorable SNPs in the population (0, 1, 2, or 3).

Results

Functional class III SNPs

Of 26 SNPs, 15 are organized into 4 haplotype blocks, and 11 are encoded outside of the blocks (Figure 1; supplemental Figure 1). Class III SNPs that merited inclusion into class I~class III~class II haplotypes were identified in cohort I patients and donors by interrogating each block (genotype strings, phased haplotypes, and block matching) and each single SNP outside of blocks (Table 2; supplemental Tables 2 and 3).

Mismatching for haplotype blocks or single SNPs did not show any definitive association with outcome in cohort I and was not tested in cohort II (supplemental Table 4).

Cohort I SNP blocks

In patients, SNP-3~4~5~6~7~8~9~10 and SNP-23~25 blocks showed evidence of association with outcome; however, no single SNP within either block was definitively associated with outcome (supplemental Tables 2 and 3). In donors, SNP-3~4~5~6~7~8~9~10, SNP-11~12~13, and SNP-14~15 blocks showed evidence of association with outcome, and the testing of individual SNPs showed evidence of associations with donor SNP3, SNP11, and SNP14 with outcome (Table 2).

Cohort I single SNPs that reside outside of blocks

The 11 SNPs were individually tested for genotype and mismatch associations with outcome in patients separately from donors (Table 2). In patients, SNP2 (rs915654) was associated with both mortality and relapse, with the A allele being protective relative to T; SNP1 was definitively associated only with relapse. In donors, SNP2 and SNP20 were definitively associated with both mortality and relapse, whereas SNP1, SNP3, SNP11, SNP14, and SNP21 were associated only with relapse. Similar to patients, the A allele for SNP2 in donors was protective relative to T. All remaining patient and donor SNPs failed to show definitive evidence of an association with outcome (supplemental Table 3).

In summary, favorable alleles were *HLA-B*-21Thr, *HLA-E*-107Gly, rs915654A, and *HLA-DRB1*-86Gly. Unfavorable alleles were *HLA-B*-21Met, *HLA-E*-107Arg, rs915654T, and *HLA-DRB1*-86Val. Each SNP allele was consistently risky or protective for end points; no allele was risky for 1 end point but protective for a different end point.

Additive SNP-haplotype effects

To achieve the most robust definition of class I~class III~class II haplotypes, class I~class II foundational haplotypes that best

Table 2. Class III SNPs associated with mortality and/or relapse

SNP	Patients or donors	Genotype	Mortality*		Relapse*	
			No.	Hazard ratio (95% confidence interval); P value	No.	Hazard ratio (95% confidence interval); P value
SNP1 (rs928815)	Patients	GG	558	1.0	501	1.0
		GT	678	0.97 (0.81-1.17); .78	620	1.22 (1.00-1.48); .05
		TT	191	0.91 (0.70-1.20); .51	175	1.22 (0.91-1.63); .18
	Donors	GG	581	1.0	520	1.0
		GT	648	0.94 (0.78-1.13); .51	595	1.04 (0.85-1.27); .69
		TT	197	1.12 (0.87-1.45); .39	181	1.36 (1.03-1.78); .03
SNP2 (rs915654)	Patients	TT	449	1.0	407	1.0
		AT	639	0.75 (0.62-0.91); .004	580	0.84 (0.68-1.03); .10
		AA	249	0.80 (0.62-1.03); .08	228	0.69 (0.52-0.92); .01
	Donors	TT	449	1.0	407	1.0
		AT	639	0.75 (0.62-0.91); 0.004	580	0.84 (0.68-1.03); .10
		AA	249	0.80 (0.62-1.03); 0.08	228	0.69 (0.52-0.92); .01
SNP3 (rs2229094)	Donors	TT	na†	na†	696	1.0
		CT	na†	na†	506	0.82 (0.68-0.99); .04
		CC	na†	na†	93	0.95 (0.66-1.34); .75
SNP11 (rs2256965)	Donors	AG	na†	na†	610	1.0
		AA	na†	na†	203	1.35 (1.04-1.76); .02
		GG	na†	na†	485	1.07 (0.87-1.30); .53
SNP14 (rs1052248)	Donors	TT	na†	na†	680	1.0
		AA	na†	na†	112	0.93 (0.67-1.30); .68
		AT	na†	na†	505	0.75 (0.62-0.91); .004
SNP20 (rs755714)	Donors	CC	780	1.0	702	1.0
		CT	530	1.21 (1.01-1.46); .04	486	1.37 (1.12-1.67); .002
		TT	114	1.09 (0.79-1.52); .60	106	1.35 (0.96-1.89); .08
SNP21 (rs2075800)	Donors	GluGlu	757	1.0	680	1.0
		GluLys	536	1.17 (0.97-1.42); .09	495	1.23 (1.00-1.50); .05
		LysLys	135	1.03 (0.75-1.40); .87	123	1.48 (1.09-2.01); .01

A haplotype block approach was taken to examine the hazards of overall mortality and relapse. Each of the 4 SNP-haplotype blocks (SNP-3~4~5~6~7~8~9~10, SNP-11~12~13, SNP-14~15, and SNP-23~25) was evaluated to assess associations with the outcome. When a block was found to be associated with a clinical end point, individual SNPs within that block were individually tested for associations with that same end point. If a block did not associate with an end point, individual SNPs were not evaluated for that end point. Of the 26 SNPs, 11 reside outside of the 4 blocks (SNP1, 2, 16, 17, 18, 19, 20, 21, 22, 24, and 26) and were individually assessed for associations with transplant outcome. Summarized are SNPs that showed association with outcome (defined as $P \leq .05$ for at least 1 of mortality and relapse), whether they were identified through block or nonblock single SNP models. A hazard ratio of ≥ 1.0 for both mortality and relapse was used to define the unfavorable allele of an SNP. Unfavorable donor alleles are SNP1T (relapse), 2T (mortality and relapse), 3T (relapse); 11A (relapse), 14T (relapse), 20T (mortality and relapse), and 21LYS (mortality and relapse). Unfavorable patient alleles are 1T (relapse) and 2T (mortality and relapse).

CMV, cytomegalovirus; DRI, disease risk index; Glu, glutamic acid (rs2075800); Lys, lysine (rs2075800); na, not applicable.

*Models adjusted for year of transplant, conditioning regimen intensity, DRI, patient/donor CMV serostatus, patient age, donor age, stem cell source, patient sex, donor sex, use of total body irradiation, comorbidity index, donor relationship, patient race, and number of HLA mismatches.

†Individual SNP models were not examined for an end point if the block within which the SNP resides was not associated with the end point.

describe mortality and relapse risks were first determined in cohort I patients separately from donors using markers known to influence relapse and/or mortality in PTCy-based haploidentical HCT.^{18,19,21}

For donors, *HLA-B*-Met-21Thr~*DRB1*-Val86Gly led to a better model for mortality than *HLA-E*-Gly107Arg~*DRB1*-Val86Gly (log likelihood $P = .04$ vs $P = .19$, respectively) and to a similar model for relapse ($P = .13$ vs $P = .14$, respectively). Therefore, the former was identified as the foundational haplotype for donors. We then examined the impact of each of the 7 candidate class III donor SNPs when added to this foundational haplotype. Inclusion of SNP2 led to improved models for both mortality (log-likelihood $P = .01$) and relapse (log-likelihood $P = .004$), deemed a greater improvement than with other candidate SNPs ($P = .04$ and $P = .05$ for SNP11; $P = .03$ and $P = .007$ for SNP14; $P = .13$ and $P = .005$ for SNP20; $P = .22$ and $P = .02$ for SNP21 for mortality and relapse, respectively), leading to the selection of *HLA-B*-Met-21Thr~rs915654~*DRB1*-Val86Gly as the extended SNP haplotype in donors.

For patients, the model fit to mortality and relapse was collectively similar between *HLA-E*-Gly107Arg~*DRB1*-Val86Gly and *HLA-B*-Met-21Thr~*DRB1*-Val86Gly. Therefore, each candidate patient SNP (SNP1 and SNP2) was assessed for both haplotypes. When SNP2 was added to each foundational haplotype, the models for both mortality and relapse were improved to a larger degree than when adding SNP1 ($P = .10$ vs $P = .67$ for mortality and *HLA-E*-Gly107Arg~*DRB1*-Val86Gly; $P = .12$ vs $P = .55$ for mortality and *HLA-B*-Met-21Thr~*DRB1*-Val86Gly; $P = .06$ vs $P = .24$ for relapse and *HLA-E*-Gly107Arg~*DRB1*-Val86Gly; $P = .05$ vs $P = 0.20$ for relapse and *HLA-B*-Met-21Thr~*DRB1*-Val86Gly). Collectively, the addition of SNP2 to *HLA-E*-Gly107Arg~*DRB1*-Val86Gly led to better models than *HLA-B*-Met-21Thr~*DRB1*-Val86Gly and motivated the selection of *HLA-E*-Gly107Arg~rs915654~*DRB1*-Val86Gly as the extended SNP haplotype in patients.

Each individual encodes 6 alleles across 3 SNP positions. We tested the hypothesis that clinical outcome depends on the total number of (un)favorable SNPs (Table 3). In cohort I patients, as the total number of favorable SNPs decreased, mortality and relapse were numerically increased. In cohort I donors, there was also a numerical increase in the hazards of mortality and relapse as the total number of favorable SNPs decreased. Because the 3-SNP haplotype was identified from results observed for each marker, an association between the number of (un)favorable SNPs and outcome was expected. Therefore, we examined this same hypothesis in the validation cohort II, in which the direction of associations was the same as those in cohort I (Table 3). Formal tests of interaction examined potential statistical interactions between the number of (un)favorable patient and donor SNPs and yielded $P = .39$ for mortality and $P = .65$ for relapse. The effect of the number of (un)favorable SNPs on nonrelapse mortality in cohorts I and II (using the same methods as for relapse mortality) yielded qualitatively similar results as those for overall mortality and relapse (supplemental Table 5). These results suggest that the total number of patient and donor (un)favorable SNPs may be a determinant of transplant outcome.

Functional *HLA-A~B~DRB1* haplotypes

The aforementioned additive effects of SNPs count the number of (un)favorable alleles of the total 6 alleles at 3 SNP positions and do not consider the phase of the SNPs as 3-marker haplotypes. We tested the hypothesis that phased *HLA-E*-Gly107Arg~rs915654~*DRB1*-Val86Gly in patients and *HLA-B*-Met-21Thr~rs915654~*DRB1*-Val86Gly in donors provide additional information on mortality and relapse beyond that gleaned from 6 total alleles. In patients, 8 unique phased haplotypes included 1 haplotype with 0 favorable SNPs (Arg~T~Val), 3 with 1 favorable SNP (Gly~T~Val, Arg~A~Val, and Arg~T~Gly), 3 with 2 favorable SNPs (Gly~A~Val, Gly~T~Gly, and Arg~A~Gly), and 1 with 3 favorable SNPs (Gly~A~Gly). The hazards of mortality showed some evidence of differences across these SNP haplotypes ($P = .05$; $P = .66$ for relapse).

In donors, among the 8 *HLA-B*-Met-21Thr~rs915654~*DRB1*-Val86Gly haplotypes, 1 had 0 favorable SNP (Met~T~Val), 3 had 1 favorable SNP (Thr~T~Val, Met~A~Val, and Met~T~Gly), 3 had 2 favorable SNPs (Thr~A~Val, Thr~T~Gly, and Met~A~Gly), and 1 had 3 favorable SNPs (Thr~A~Gly). There was some evidence that the hazards of mortality ($P = .06$) and relapse ($P = .05$) differed across phased donor SNP haplotypes.

Each SNP haplotype is part of a broader *HLA-A~B~DRB1* allele haplotype. The 2770 patients and donors most commonly encoded A*01~B*08~*DRB1**03:01, A*03~B*07~*DRB1**15:01, A*02~B*44~*DRB1**04:01, and A*29~B*44~*DRB1**07:01 (Figure 2; supplemental Table 6). Notably, 2 of these 4 haplotypes (A*01~B*08~*DRB1**03:01 and A*03~B*07~*DRB1**15:01) encoded almost exclusively unfavorable patient and donor SNP haplotypes. For example, when A*01~B*08~*DRB1**03:01 is defined for donor SNPs, 99% carried Met~T~Val (0 favorable SNPs), and 1% carried Met~A~Val (1 favorable SNP). In contrast, the remaining 2 haplotypes, A*29~B*44~*DRB1**07:01 and A*02~B*44~*DRB1**04:01, more frequently encoded favorable SNP haplotypes. For example, when *HLA-A**02~B*44~*DRB1**04:01 is defined for donor SNPs, 93% carried Thr~A~Gly (3 favorable SNPs), and 7% carried Thr~T~Gly (2 favorable SNPs). When extended haplotypes are defined at higher levels of resolution and include *HLA-C* and *-DQB1*, diversity of SNP haplotypes is evident (supplemental Table 6).

Some *HLA-A~B~DRB1* haplotypes encoded a diversity of SNP haplotypes. *HLA-A2~B35~DR4* is among the most diverse haplotypes and can encode any of the 8 *HLA-E*-Gly107Arg~rs915654~*DRB1*-Val86Gly combinations because of variation at *HLA-E*-Gly107Arg, rs915654A/T, and *HLA-DRB1*-Val86Gly, the latter defined by the *DRB1**04 allele subtype (supplemental Table 6). Finally, testing of *HLA-E*-Gly107Arg and rs915654 in a panel of 102 HLA homozygous cell lines confirmed the haplotype content observed in patients and donors (supplemental Table 7).

Clinical utility of haplotypes for donor selection

The findings suggest that higher numbers of favorable SNPs in the donor are associated with better outcomes in the patient. We explored whether the expected number of favorable SNPs

Table 3. Effect of the number of favorable SNPs on outcome

End point	Total no. of favorable SNPs		Cohort I		Cohort II	
			n	Adjusted HR	n	Adjusted HR
Mortality	Patient	6	14	1	11	1
		5	97	0.79	83	1.24
		4	226	0.95	224	1.13
		3	398	1.06	345	1.20
		2	360	1.07	283	1.49
		1	162	1.37	140	1.65
		0	41	1.66	30	1.45
		Trend test	1298	Adjusted HR, 1.12 (95% CI, 1.04-1.20); $P = .002$	1116	Adjusted HR, 1.10 (95% CI, 1.01-1.20); $P = .03$
	Donor	6	57	1	51	1
		5	216	1.09	178	1.04
		4	351	0.90	300	1.12
		3	376	1.34	299	1.12
		2	214	1.22	184	1.24
		1	78	1.45	84	1.39
		0	25	0.73	19	1.54
		Trend test	1317	Adjusted HR, 1.06 (95% CI, 0.99-1.14); $P = .08$	1115	Adjusted HR, 1.06 (95% CI, 0.99-1.15); $P = .11$
Relapse	Patient	6	13	1	10	1
		5	90	1.44	74	1.70
		4	204	1.57	207	0.88
		3	353	1.47	318	1.25
		2	333	1.59	258	1.34
		1	151	1.72	132	1.46
		0	37	2.06	24	1.67
		Trend test	1181	Adjusted HR, 1.05 (95% CI, 0.97-1.14); $P = .19$	1116	Adjusted HR, 1.10 (95% CI, 1.01-1.20); $P = .03$
	Donor	6	52	1	47	1
		5	201	1.33	163	1.21
		4	315	1.45	273	1.21
		3	341	1.37	280	1.14
		2	193	1.54	165	1.26
		1	76	2.28	77	0.97
		0	24	0.69	18	2.15
		Trend test	1202	Adjusted HR, 1.06 (95% CI, 0.98-1.13); $P = .13$	1115	Adjusted HR, 1.06 (95% CI, 0.99-1.15); $P = .11$

Multivariable models for overall mortality and relapse according to the total number of favorable SNPs in patients and donors in cohort I and cohort II. Trend tests measure hazards for the clinical end point in cohort I and II patients and donors. Models adjusted for year of transplant, conditioning regimen intensity, disease risk index, patient/donor cytomegalovirus serostatus, patient age, donor age, stem cell source, patient sex, donor sex, use of total body irradiation, comorbidity index, donor relationship, patient race, and number of HLA mismatches.

CI, confidence interval; HR, hazard ratio.

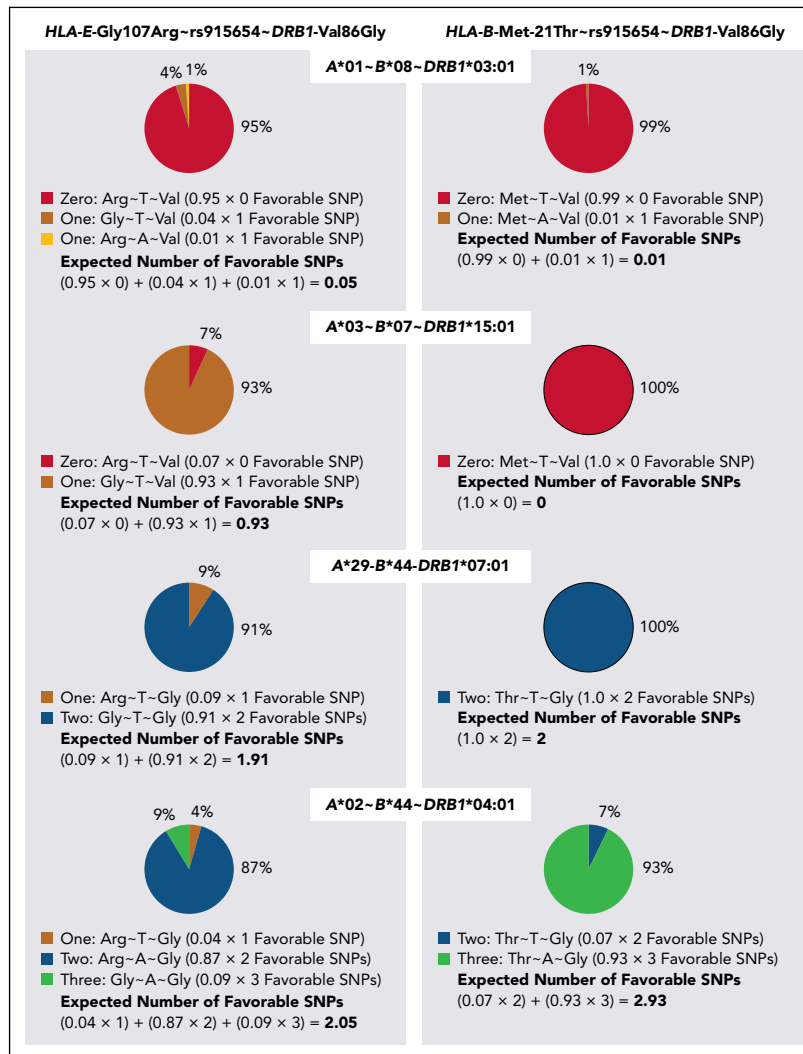


Figure 2. The expected number of favorable SNPs for 4 common HLA-A-B-DRB1 haplotypes. Frequencies of the 4 most commonly observed haplotypes phased for both HLA-A-B-DRB1 and 3-SNP haplotypes in 2770 haploidentical transplants are shown (the shared haplotype was counted once). These 4 HLA-A-B-DRB1 haplotypes are defined by low-resolution typing of HLA-A and HLA-B and by high-resolution typing of HLA-DRB1. A total of 3592 haplotypes were phased for both HLA-A-B-DRB1 and HLA-E-Gly107Arg-rs915654~DRB1-Val86Gly (rs1264457~rs915654~rs17885482), and 4570 were phased for both HLA-A-B-DRB1 and HLA-B-Met-21Thr-rs915654~DRB1-Val86Gly (rs1050458~rs915654~rs17885482). The haplotypes are A*01~B*08~DRB1*03:01 (n = 84 patients and 106 donor haplotypes), A*03~B*07~DRB1*15:01 (n = 44 patients and 57 donor haplotypes), A*29~B*44~DRB1*07:01 (n = 22 patients and 35 donor haplotypes), and A*02~B*44~DRB1*04:01 (n = 23 patients and 28 donor haplotypes). Each HLA-A-B-DRB1 is defined by HLA-E-Gly107Arg-rs915654~DRB1-Val86Gly SNP markers (left; informative for patients) and by HLA-B-Met-21Thr-rs915654~DRB1-Val86Gly SNP markers (right; informative for donors). For each HLA-A-B-DRB1 thus defined, the frequencies of the respective observed 3-SNP marker haplotypes are shown. For example, when all A*01~B*08~DRB1*03:01 haplotypes in the study population are defined by HLA-E-Gly107Arg-rs915654~DRB1-Val86Gly, 95% of chromosomes carry Arg~T~Val (0 favorable SNP), 4% carry Gly~T~Val (1 favorable SNP), and 1% carry Arg~A~Val (1 favorable SNP). These distributions are used to estimate the total expected number of favorable SNPs for the A*01~B*08~DRB1*03:01 haplotype, which is the sum of the product of the frequency of each 3-SNP marker haplotype that was observed in individuals in the study population who carried an A*01~B*08~DRB1*03:01 haplotype and the associated number of favorable SNPs. In this way, the expected number of favorable SNPs per HLA-A-B-DRB1 haplotype is not necessarily an integer because it can be interpreted as an expected value or a weighted average of the number of favorable SNPs, with the weights taken from the frequency of the relevant SNP. Using the same example for A*01~B*08~DRB1*03:01, the expected number of favorable SNPs is the sum of the product of observed frequencies and number of favorable SNPs, namely $(0.95 \times 0 \text{ [for Arg~T~Val]}) + (0.04 \times 1 \text{ [for Gly~T~Val]}) + (0.01 \times 1 \text{ [for Arg~A~Val]})$, or 0.05 expected favorable SNPs. Finally, for a given diploid individual, the total expected number of favorable SNPs is the sum of the scores for the 2 haplotypes. A summary of phased haplotypes defined at high resolution for HLA-A, -B, and -DRB1, and, when available, HLA-C and -DQB1, is provided in supplemental Table 6.

in donors could be predicted from their HLA-A, -B, and -DRB1 tissue types; if this were to be feasible, then tissue type could be used to prioritize donors with the largest expected number of favorable SNPs. Because HLA-B-Met-21Thr and HLA-DRB1-Val86Gly are defined directly from the HLA-B and HLA-DRB1 tissue type, respectively,^{17,18} we tested the hypothesis that the donor's HLA-A-B-DRB1 haplotypes are

surrogates for favorable SNPs contributed by HLA-B-Met-21Thr-rs915654~DRB1-Val86Gly haplotypes.

Figure 3A illustrates the representation of a SNP haplotype to its parental HLA-A-B-DRB1 haplotype. We discovered that each HLA-A-B-DRB1 haplotype has a characteristic number of favorable SNPs (supplemental Table 6). With knowledge of the

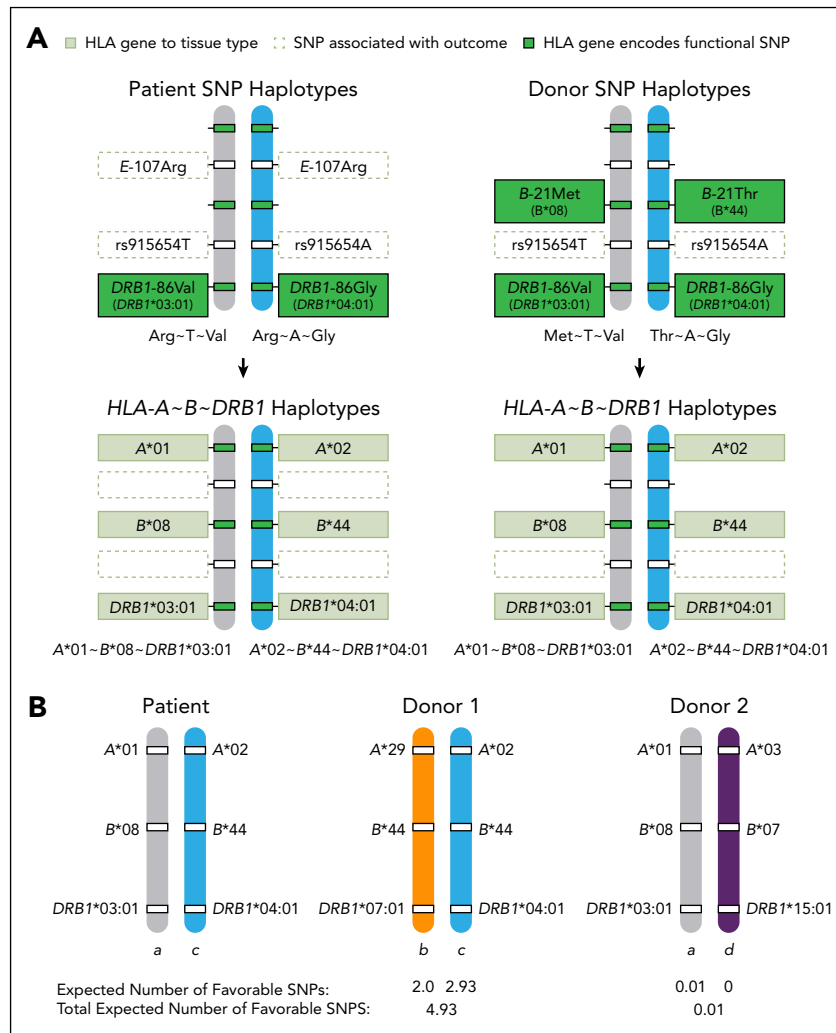


Figure 3. HLA-A~B~DRB1 haplotypes and their SNP proxies for the selection of candidate haploidentical donors. (A) Each diploid individual carries 2 parental haplotypes that are illustrated by the gray and blue chromosomes. Each parental haplotype carries triads of SNP markers that have functional significance in patients and in donors. The risk of mortality is influenced by 3 SNP markers in patients (HLA-E-Gly107Arg, rs915654, and HLA-DRB1-Val86Gly) and by 3 SNP markers in donors (HLA-B-Met-21Thr, rs915654, and HLA-DRB1-Val86Gly). Two representative parental haplotypes, A*01~B*08~DRB1*03:01 and A*02~B*44~DRB1*04:01, show how a given haplotype can be defined by either the 3 relevant patient SNPs (left) or by the 3 relevant donor SNPs (right). In a true haploidentical patient-donor pair, HLA-E-Gly107Arg~rs915654~DRB1-Val86Gly and HLA-B-Met-21Thr~rs915654~DRB1-Val86Gly phase can be established because the patient and haploidentical donor, by definition, must share 1 complete haplotype. Whereas HLA-E-Gly107Arg and rs915654 are functional polymorphisms not currently tested in clinical practice, HLA-B-Met-21Thr and HLA-DRB1-Val86Gly are sequence features that are specified by the HLA-B and HLA-DRB1 tissue type, respectively, which are routinely ascertained before transplant. By virtue of their physical linkage on the same chromosome, each 3-SNP haplotype can be represented by its associated HLA-A~B~DRB1 haplotype (down arrows). (B) A patient's HLA haplotypes cannot be modified; however, patients who have ≥2 haploidentical family members may have a choice in donors. A candidate haploidentical donor with a higher total expected number of favorable SNPs might be considered over a donor with a lower expected number of favorable SNPs. For a given donor haplotype, the total expected number of favorable SNPs is the sum of the product of the frequencies of favorable SNPs contributed by each HLA-B-Met-21Thr~rs915654~DRB1-Val86Gly haplotype and the associated number of favorable SNPs. Shown are 2 candidate haploidentical donors and 4 parental haplotypes. The parental haplotypes with their corresponding expected number of favorable SNPs are the same haplotypes displayed in Figure 2 (A*29~B*44~DRB1*07:01 [orange], A*02~B*44~DRB1*04:01 [blue], A*01~B*08~DRB1*03:01 [gray], and A*03~B*07~DRB1*15:01 [purple]). The total expected number of favorable SNPs in donor 1 (4.93) is higher than that of donor 2 (0.01), and donor 1 could therefore be prioritized. In donors, the total expected number of favorable SNPs is captured by the donor's HLA-A, -B, and -DRB1 tissue types and does not require direct typing of rs915654.

donor's 2 parental haplotypes (determined through pretransplant pedigree HLA typing), the expected number of favorable SNPs associated with each donor HLA-A~B~DRB1 haplotype can be calculated (Figure 3B), and the sum of both haplotypes yields the total expected number of favorable SNPs. For patients who have a choice of haploidentical family members, HLA-A~B~DRB1 haplotypes serve as proxies for favorable

SNPs and can be leveraged to prioritize donors with a higher expected number of favorable SNPs. The choice of HLA-A, HLA-B, and HLA-DRB1 to define classic 3-locus HLA haplotypes was attractive because it would not only parallel the extensive population data gleaned from disease-association studies⁵⁻⁷ but would be clinically actionable and obviate the need to test rs915654 directly in donors.

Discussion

Major advances in transplantation have contributed to improved survival of patients over time, but relapse remains the primary reason for transplant failure.^{27,28} Current understanding of the HLA barrier includes the risks associated with single genes and gene features; however, an emerging paradigm involves the synergistic effects conferred by multiple co-inherited loci that are best captured as haplotypes.¹⁸ Although class I and II transplantation determinants are well-known, a major limitation is the lack of data for class III that are required to bridge the 3 major regions of the MHC. This study narrowed this gap by identifying functional class III polymorphisms that show evidence of influencing mortality and relapse after HCT and how such variants synergize with class I and II loci as extended haplotypes. The same HLA haplotypes associated with lower survival after HCT are also strongly linked to autoimmune diseases.⁵⁻⁷ These findings have far-reaching implications for understanding the shared biological processes across distinct human diseases. The mapping of *HLA-E*, *HLA-B*, rs915654, and *HLA-DRB1* variants to their classical HLA haplotypes reinforces the high-risk nature of several well-known common extended haplotypes, notably *HLA-A1~B8~DR3* and *HLA-A3~B7~DR15*, and extends knowledge of the sentinel role of HLA haplotypes in human diseases.^{6,7} That both inflammatory haplotypes are also high risk in transplantation opens entirely new avenues in risk assessment and in the selection of candidate haploidentical donors.

To surmount the potential challenges to disease mapping posed by LD, we applied classic principles of haplotype mapping supplemented with single SNP analyses to identify functional class III SNPs. The reliable pedigree analysis of haploidentical patient/donor pairs secured phased genomic haplotypes across class I~class III~class II. Functional extended haplotypes were built from 2-locus class I~class II foundational haplotypes comprised variants previously shown to affect mortality and relapse.^{18,19,21} The addition of the candidate class III variants to foundational haplotypes permitted the identification of extended class I~class III~class II haplotypes that inform relapse and mortality. Haplotypes that span class I~class III~class II provide a broad, comprehensive survey of MHC region variation. The functional genetic features were SNP genotypes and haplotypes and not patient/donor (mis)matching. As such, the definition of patient haplotypes was not the same as that for donors. The potential cumulative effects of haplotype-linked markers on mortality and relapse support a paradigm based on the concerted effects of germ line variation more so than (mis)matching. The consistency of results between cohorts I and II strengthened the notion that haplotypes may be an integral part of the functional transplantation barrier and that variants encoded across long genetic distances might participate in alloimmune responses.

Extensive experience in mapping causative genes for HLA-associated autoimmune disorders has highlighted the importance of noncoding variation in disease susceptibility^{29,30} and motivated us to study class III variants that influence gene expression. Our SNP panel included expression quantitative trait loci (eQTLs) in addition to missense variants descriptive of world populations, but other forms of functional variation might contribute to mortality and relapse as well, for example, loci within the centromeric class III region, which were not tested.⁸

Therefore, the 3-SNP haplotypes in this study may be considered one definition of functional haplotypes, not necessarily the only definition. Furthermore, although the 2 cohorts provided >8100 phased haplotypes, haplotype variation across human populations is significant.^{8,31} A comprehensive analysis of population-based haplotypes was not feasible because of the limited number of non-White families in this study and remains an important question. Multivariable models adjusted for nongenetic factors known to influence mortality and relapse; however, the lack of *HLA-DPB1* typing precluded meaningful analysis and its inclusion into models. Future studies of sufficient size and diversity are also of interest to confirm that a risky variant(s) that is shared among diverse haplotypes confers similar effects.³²

The biological significance of SNP2, rs915654, might involve *LST1* and its expression. rs915654 is located upstream of *LTA* and functions as an eQTL in whole blood for *LST1* and as a biomarker in rheumatoid arthritis.^{33,34} It is noteworthy that rs915654 was associated with outcome in both patients and donors, and 2 additional candidate donor SNPs were functionally related to *LST1* expression (SNP11 and SNP14).^{33,35} These data support a role for *LST1* expression in clinical outcome in haploidentical HCT, but the precise mechanisms remain to be explored in future studies. rs915654 is likely playing a major role in the outcome because it is not in LD with any other marker across 2 megabases of the MHC (supplemental Figure 2). *LST1* transcripts are constitutively expressed in peripheral mononuclear cells, T cells, B cells, and dendritic cells. They may inhibit lymphocyte proliferation and cell differentiation and induce apoptosis.³⁶⁻³⁸ The rs915654-associated *LST1* expression correlates with inflammatory and autoimmune diseases.^{32,39} The most direct evidence for a role for *LST1* is in acute myeloid leukemia, in which high *LST1* expression is associated with poor clinical outcome.⁴⁰ The precise mechanisms by which *LST1* exerts its inhibitory signaling are not entirely known, but potential roles for soluble protein, coexpression of neighboring genes, and T-cell immunoglobulin and mucin-domain containing-3-mediated cell death and T-cell exhaustion remain important mechanisms for future studies.^{41,42}

It is not necessary to prospectively type *HLA-E* (patients) or rs915654 (patients and donors) to derive the clinical benefits from SNPs. Knowledge of *HLA-A*, *-B*, and *-DRB1* is sufficient to predict the number of favorable donor SNPs at least for common haplotypes and is a particularly attractive approach for donor selection, strengthened by the availability of worldwide haplotype variation.³¹ Routine typing of *HLA-B* and *HLA-DRB1* in donors directly informs the leader and residue 86 dimorphisms, respectively.^{17,18} For patients with ≥ 2 otherwise equivalent haploidentical family members, consideration of a donor with a high number of favorable SNPs may supplement conventional selection criteria (Figure 3). With respect to unrelated donors and cord blood units, >43 million typed donors and units are currently available for patient care.⁴³ In the absence of family studies, haplotype estimation tools are readily available and already incorporated into donor search approaches and can be used to define haplotypes from which the number of favorable SNPs can be estimated.⁴⁴ Transplantation of donors or units whose HLA haplotypes encode a higher content of favorable SNPs would be predicted to result in better patient outcomes than those with mainly unfavorable SNPs and offers an individualized approach to the patient's transplant. Importantly, because the HLA haplotype paradigm

does not depend on donor matching, it can be readily incorporated alongside current match-based algorithms.

In summary, HLA haplotype-associated genomic variation may be a key feature of the transplantation barrier. New information on functional haplotypes has the potential to improve transplant survivorship and extend knowledge of the biology of HLA-associated diseases. Because donor *HLA-B*-Met-21Thr~rs915654~*DRB1*-Val86Gly haplotypes and functional SNPs are proxies for one another, knowledge of donor *HLA-A~B~DRB1* haplotypes alone can be used to prioritize the selection of candidate donors associated with optimal survival outcomes for future patients.

Acknowledgments

The authors thank Dawn Miller and Mark Gatterman for outstanding technical assistance.

This study was supported by grants from the National Institutes of Health: AI069197 from the National Institute of Allergy and Infectious Diseases (E.W.P., T.G., M.M., C.M., P.S., and S.R.S.); CA218285 from the National Cancer Institute (E.W.P., T.G., M.M., P.S., and C.M.); CA100019 from the National Cancer Institute (E.W.P., T.G., M.M., C.M., and P.S.); and CA231838 from the National Cancer Institute (E.W.P., T.G., M.M., C.M., and P.S.). The Center for International Blood and Marrow Transplant Research (CIBMTR) is supported primarily by the Public Health Service U24CA076518 from the National Cancer Institute, the National Heart, Lung, and Blood Institute, and the National Institute of Allergy and Infectious Diseases; 75R60222C00011 from the Health Resources and Services Administration; and N00014-24-1-2057 and N00014-25-1-2456 from the Office of Naval Research. Support for CIBMTR is also provided by the Medical College of Wisconsin, National Marrow Donor Program, Gateway for Cancer Research, and the Pediatric Transplantation and Cellular Therapy Consortium.

Authorship

Contribution: E.W.P. designed the study and drafted the manuscript; P.S. and T.G. performed statistical analysis; and all authors assembled the data, critically reviewed and edited the manuscript, and approved the final version.

Conflict-of-interest disclosure: E.W.P., T.G., C.M., M.M., P.S., and S.R.S. report grants from the National Institutes of Health. S.R.S. reports grants from the US Office of Naval Research. M.H. declares no competing financial interests.

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Footnotes

Submitted 5 January 2026; accepted 8 April 2026; prepublished online on *Blood* First Edition 22 April 2026. <https://doi.org/10.1182/blood.2025033031>.

Data are available from the corresponding author, Effie W. Petersdorf (epetersd@fredhutch.org), on request. Data sharing for research purposes may have partial restrictions consistent with the informed consent of study participants from whom the data were obtained.

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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